

Associator Field Theory

A Research Manuscript on One-Bug Magmas, Cayley–Dickson Curvature, and Associator Sheaves

Version 2: consolidated through the Associator Atlas Compression and Marginal Chart Laws

Purpose

This document consolidates discoveries, proof sketches, computations, equations, and applications concerning finite magmas, nonassociativity, and Cayley–Dickson associator curvature. Claims are labeled by status: proved by construction or proof sketch, computationally certified in tested ranges, or conjectural.

Executive Summary

- Finite magmas are sets with one binary operation and no assumed laws. One target is the study of one-bug magmas with large semigroup repair distance.
- Explicit one-bug families were constructed. The compact fan family F_m has $2m + 4$ elements, $\text{ns}(F_m) = 1$, and $\text{sdist}(F_m) = m + 1$.
- Small hard examples were found computationally, including six- and eight-element one-bug magmas of semigroup distance 3 and 5.
- A hidden-state interpretation emerged: many hard one-bug examples are bad quotients of associative systems. Adding one missing state can collapse repair distance from large to one.
- The work then shifted to Cayley–Dickson basis-unit loops, which model the progression from complex numbers to quaternions, octonions, sedenions, and beyond. Their associators are not missing-state errors but structured cocycle curvature.
- A theory of associator fields, source separation, local potentials, gluing kernels, holonomy spectra, coherence automata, atlases, and compression laws was built.
- The most useful practical law is: diagnose the source of nonassociativity before repair. It may be table error, a missing state, cocycle or coherence curvature, or a sheaf-like gluing obstruction.

Core object	Meaning	Main equation or law
Associator field	Raw failure of associativity	$\alpha_\mu(x, y, z) = e_{\mu(\mu(x, y), z)} - e_{\mu(x, \mu(y, z))}$
Projection curvature	Failure caused by hidden-state compression	$\alpha_o = R_\pi - L_\pi$
Cocycle curvature	Failure caused by a twist or phase law	$\Phi = \delta f$
Local potential	Three-plane description of Cayley–Dickson curvature	$\Phi _P = \Delta_P(1 + \theta_P(u) + \theta_P(w))$
Gluing kernel	Pairwise compatibility of local potentials	$K(0) = 1, K(1) = 101/155, K(2) = 3/5, K(3) = 1$
Holonomy spectrum	Single-gauge coherence profile	$H_d(z)$ recurrence and closed spectrum
Atlas number	Minimum charts needed for exact coherence	$\tau_d = d - 2$
Compression curve	Structured approximate coherence	$F_r = 1 - \left(\frac{7}{2^{r+2}} - \frac{14}{4^{r+2}} + \frac{1}{8^{r+1}} \right)$

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1 Motivation and Research Stance

The aim is to study a blind spot in mathematics where simple objects hide large unknown structure. Finite magmas are ideal for this purpose. A magma is merely a set with a binary operation; there is no identity, inverse, associativity, commutativity, or distributivity unless imposed. This weakness creates a vast combinatorial universe that classical algebra usually avoids by focusing on highly structured islands such as groups, rings, fields, lattices, loops, semigroups, and quasigroups.

The guiding question is not only how much associativity fails, but what kind of thing an associativity failure is. The analysis repeatedly shows that a scalar failure count is insufficient. The same visible associator can arise from a wrong table entry, a missing hidden state, a quotient of an associative system, or coherent cocycle curvature like the octonions.

Working principle. Do not repair a nonassociative operation until its associator source has been diagnosed: patch error, split missing state, or track coherence curvature.

2 Definitions and Baseline Measures

Let A be a set with binary operation $*$. Its associativity defect at a triple (x, y, z) is the comparison between $(x * y) * z$ and $x * (y * z)$. Define

$$\text{ns}(A) = \#\{(x, y, z) \in A^3 : (x * y) * z \neq x * (y * z)\}.$$

The semigroup distance $\text{sdist}(A)$ is the minimum number of multiplication-table entries that must be changed to make the operation associative:

$$\text{sdist}(A) = \min\{\text{table-entry Hamming distance from } A \text{ to an associative operation}\}.$$

A one-bug magma is a finite magma with $\text{ns}(A) = 1$. In the groupoid literature these are often called Szasz–Hajek groupoids or SH-groupoids. One core problem is

$$D_n = \max\{\text{sdist}(A) : |A| = n \text{ and } \text{ns}(A) = 1\}.$$

Measure	What it sees	What it misses
$\text{ns}(A)$	Visible local associativity failures	Repair cost, hidden-state explanations, coherent curvature
$\text{sdist}(A)$	Internal table repair cost	Whether adding states would repair cheaply
Associative spectrum	How many bracketings induce distinct n -ary operations	Which entries must be repaired
Projection rank / hidden-state cost	How many missing states explain the defect	Internal repair difficulty
Holonomy / atlas complexity	Whether local repairs glue globally	Raw local failure count

3 One-Bug Finite Magmas and Semigroup Distance

3.1 Early computational reconnaissance

Brute-force enumeration for $n \leq 3$ and targeted exact search for later examples established a small-order baseline. The key lesson was that very small one-bug magmas are not yet globally entangled: at order 3, every one-bug example is one edit from a semigroup. Later exact searches found the first hard small examples.

Order n	Result
2	No one-bug magmas found.
3	60 one-bug magmas; all have semigroup distance 1.
4	2,976 one-bug magmas; all found one edit from associative.
5	240,840 one-bug magmas; all found one edit from associative.
6	A one-bug magma with semigroup distance 3 exists.
8	A one-bug magma with semigroup distance 5 exists.

3.2 The compact fan family F_m

The most compact infinite family retained from the construction search is F_m . It proves a linear lower bound for internal repair distance among one-bug magmas:

$$F_m = \{0, a, b, c\} \cup \{x_i, u_i : 1 \leq i \leq m\}, \quad |F_m| = 2m + 4.$$

All unspecified products are 0. The nonzero products are

$$\begin{aligned} aa &= b, & ab &= c, \\ ax_i &= u_i, & bx_i &= c, & au_i &= c, & x_i a &= b, & u_i a &= c & (1 \leq i \leq m), \\ x_i u_j &= c & (1 \leq i, j \leq m). \end{aligned}$$

The only failed associativity triple is (a, a, a) :

$$(aa)a = ba = 0, \quad a(aa) = ab = c.$$

Thus $\text{ns}(F_m) = 1$. An explicit repair uses $m + 1$ changes: set $aa := 0$ and $au_i := 0$ for all i . A disjoint-witness argument shows no associative repair can use fewer than $m + 1$ edits. Therefore

$$\text{ns}(F_m) = 1, \quad \text{sdist}(F_m) = m + 1, \quad |F_m| = 2m + 4.$$

Status. Proved by explicit construction and disjoint witness lower bound; computationally checked for small m .

3.3 Six- and eight-element hard examples

A six-element example G has $\text{ns}(G) = 1$ and $\text{sdist}(G) = 3$. Its important mechanism is defect shifting: a one-entry patch can move the unique associativity failure to a nearby triple instead of eliminating it.

$$G = \{0, a, c, x, b, y\}.$$

Its nonzero products are

$$\begin{aligned} aa = b, \quad ax = b, \quad ab = c, \quad ay = c, \\ xa = y, \quad xb = c, \quad ba = c, \quad ya = c, \quad yx = c. \end{aligned}$$

The unique bad triple is (a, a, x) .

An eight-element example A has $\text{ns}(A) = 1$ and $\text{sdist}(A) = 5$. It was found inside a two-layer automaton-like model with letters, states, a zero, and a terminal c . Exact branch-and-bound found no associative repair through radius 4 and repairs at radius 5:

$$A = \{0, c, a, b, x, p, q, r\}.$$

The unique bad triple is (x, x, a) , since

$$(xx)a = 0, \quad x(xa) = xp = c.$$

Computationally certified:

$$\text{sdist}(A) = 5.$$

Status. The six- and eight-element distances were computationally certified by exhaustive radius search over repairs, with branch-and-bound pruning. They should be independently verified before publication.

3.4 Repair paths and energy barriers

Define visible associator energy by

$$E(T) = \text{ns}(T).$$

For the eight-element example A , every shortest 5-edit repair must temporarily pass through tables with at least three associativity failures. There is no repair path from A to a semigroup that keeps $\text{ns} \leq 2$ throughout; a path with maximum $\text{ns} = 3$ exists. Hence

$$\text{energy barrier}(A) = 3.$$

This models a real repair phenomenon: the globally correct fix may temporarily break more local tests before the system becomes coherent.

4 Hidden States and the Projection Associator Equation

4.1 Projection Associator Equation

Let $(S, \cdot)$ be associative, and let $\pi : S \rightarrow A$ be an idempotent projection onto visible states A . Define a compressed operation on A by

$$x \circ y = \pi(x \cdot y).$$

The compressed operation can be nonassociative even though S is associative. Its associator is exactly the projection curvature:

$$\alpha_\circ(x, y, z) = R_\pi(xy, z) - L_\pi(x, yz),$$

where

$$R_\pi(s, z) = e_{\pi(\pi(s) \cdot z)} - e_{\pi(s \cdot z)}, \quad L_\pi(x, t) = e_{\pi(x \cdot \pi(t))} - e_{\pi(x \cdot t)}.$$

This equation is derived algebraically by inserting the associative middle term $\pi((xy)z) = \pi(x(yz))$. It explains why hard one-bug examples can collapse after adding one hidden state.

Law. Associativity failure caused by state compression is exactly the mismatch between projection and left/right multiplication.

4.2 Hidden-state collapse

The eight-element hard table A becomes one edit from associative after adding one tailored hidden state h . The state h behaves like 0 except $ha = c$. The repair is simply $xx := h$. Thus internal repair distance 5 collapses to hidden-state repair distance 1:

$$\text{lift-sdist}_0(A) = 5, \quad \text{lift-sdist}_1(A) = 1.$$

The compact fan family F_m also collapses: although $\text{sdist}(F_m) = m + 1$, adding one hidden state h and replacing aa by h makes the system associative after one edit. This shows internal semigroup distance and hidden-state completion number can diverge arbitrarily:

$$\text{sdist}(F_m) = m + 1, \quad \text{completion}(F_m) = 1.$$

5 Transfinite Fan Theorem

The fan construction extends to an arbitrary index set I . For infinite cardinal $\kappa = |I|$, the magma F_I has exactly one associativity failure and semigroup distance κ . Since a multiplication table on κ elements has $\kappa^2 = \kappa$ entries, this is maximal in cardinal-valued distance:

$$\text{for every infinite cardinal } \kappa, \quad D_\kappa = \kappa.$$

This is a transfinite theorem, but with an important caution: infinite cardinal arithmetic hides density. Cardinality, one defect and half of all possible defects can both have size κ in infinite systems. Density along finite-dimensional approximations is needed to distinguish sparse quotient defects from dense curvature.

Status. Proved by explicit transfinite fan construction and cardinal arithmetic. It does not settle the finite quadratic-density question.

6 Cayley–Dickson Associator Curvature

6.1 Signed basis-unit loops

For the Cayley–Dickson sequence, the signed basis-unit loop is

$$Q_n = \{\pm e_u : u \in \mathbb{F}_2^n\}.$$

The familiar cases are real signs, complex units, quaternion units, octonion units, and sedenion units. Multiplication has the form

$$e_u e_v = (-1)^{f_n(u,v)} e_{u+v}.$$

Associativity is controlled by the Boolean coboundary

$$\Phi_n(u, v, w) = f_n(u, v) + f_n(u + v, w) + f_n(v, w) + f_n(u, v + w) \pmod{2}.$$

Equivalently,

$$(e_u e_v) e_w = e_u (e_v e_w) \iff \Phi_n(u, v, w) = 0.$$

6.2 Failure count law

For positive basis labels, computations and the later three-plane analysis support the exact formula

$$B_n = (2^n - 2)(2^n - 4)(2^{n-1} + 3), \quad n \geq 3.$$

For signed loops, signs do not affect whether associativity holds, so

$$\text{ns}(Q_n) = 8(2^n - 2)(2^n - 4)(2^{n-1} + 3).$$

n	Object	Signed bad triples $\text{ns}(Q_n)$	Failure density
1	Complex units	0	0
2	Quaternion units	0	0
3	Octonion units	1,344	21/64
4	Sedenion units	14,784	231/512
5	32-dimensional CD units	127,680	1995/4096
6	64-dimensional CD units	1,041,600	16275/32768

The density tends to 1/2. Thus after octonions, Cayley–Dickson nonassociativity becomes statistically dense rather than sparse.

6.3 Curvature spectrum law

Writing Φ_n as a Boolean polynomial in the bits of u, v, w , the degree spectrum appears to be

$$\text{CurvSpec}_n(t) = \sum_{d=3}^n (3^{d-1} - 3) \binom{n}{d} t^d.$$

Interpretation: octonions are pure cubic curvature; sedenions add quartic curvature; higher Cayley–Dickson algebras accumulate curvature in every degree 3 through n .

Status. Computationally verified through $n = 10$ using recurrence propagation. A proof is strongly suggested by the recurrence but not fully formalized here.

6.4 Cohomological repair distance

Restricting repairs to sign twists that preserve the quotient law $e_u e_v = \pm e_{u+v}$, associativity is the condition $\delta f = 0$. This gives a nearest-cocycle problem over $\mathbb{F}_2$:

$$d_{\text{cocycle}}(n) = \min_{\delta g=0} \#\{(u, v) : f_n(u, v) \neq g(u, v)\}.$$

n	Object	Nearest 2-cocycle distance	Status
1	Complex	0	exact
2	Quaternions	0	exact
3	Octonions	16	exact enumeration
4	Sedenions	84	exact enumeration
5	Next CD stage	≤ 372	local-search upper bound only

7 Associator Potential Theory

7.1 Three-plane potential law

Let P be a three-dimensional subspace of $\mathbb{F}_2^n$. Let $\Delta_P(u, v, w) = 1$ when u, v, w are an ordered basis of P and 0 otherwise. The discovered local law is

$$\Phi_n|_P = \Delta_P(u, v, w)(1 + \theta_P(u) + \theta_P(w)),$$

where $\theta_P : P \rightarrow \mathbb{F}_2$ is a unique linear functional. If $\theta_P = 0$, every ordered basis of P is nonassociative: the plane is octonionic. If $\theta_P \neq 0$, exactly 72 of the 168 ordered bases fail, giving curvature density $3/7$: the plane is quasioctonionic.

$$K_{\text{plane}}(P) \in \left\{1, \frac{3}{7}\right\}.$$

7.2 Plane counts

The number of octonionic and quasioctonionic three-planes in $\mathbb{F}_2^n$ appears to be

$$O_n = \begin{bmatrix} n-1 \\ 2 \end{bmatrix}_2 + \begin{bmatrix} n-1 \\ 3 \end{bmatrix}_2, \quad Q_n = 7 \begin{bmatrix} n-1 \\ 3 \end{bmatrix}_2.$$

Equivalently, with $N = 2^n$,

$$O_n = \frac{(N-2)(N-4)(N+48)}{1344}, \quad Q_n = \frac{(N-2)(N-4)(N-8)}{192}.$$

7.3 Gauge law under doubling

Under Cayley–Dickson doubling, a graph lift by a linear functional ℓ shifts the local potential by ℓ :

$$\theta_{\text{graph}(\ell)} = \theta_P + \ell.$$

Every lower three-plane has exactly one octonionic graph lift and seven quasioctonionic graph lifts. This is the local gauge mechanism behind the later sheaf and atlas laws.

8 Associator Sheaf, Holonomy, and Atlas Theory

8.1 Four-space holonomy

Inside a four-dimensional space W , there are 15 three-dimensional faces. For a global potential $\lambda \in W^*$, let $N_W(\lambda)$ count how many local three-face potentials it explains. The holonomy polynomial is

$$H_W(z) = \sum_{\lambda \in W^*} z^{N_W(\lambda)}.$$

For every tested four-space, and by the emerging structural law,

$$H_4(z) = 2z^8 + 7z^2 + 7.$$

No one gauge explains all 15 faces. The best explains 8; seven potentials explain none.

8.2 General holonomy recurrence

For d -dimensional W , the holonomy polynomial appears to depend only on d :

$$H_3(z) = z + 7,$$

$$H_d(z) = H_{d-1}(z^8) + z^{A_d} + (2^{d-1} - 1)z^{B_d}, \quad d \geq 4,$$

where

$$A_d = \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} d-1 \\ 2 \end{bmatrix}_2, \quad B_d = \begin{bmatrix} d-1 \\ 3 \end{bmatrix}_2 + \begin{bmatrix} d-2 \\ 2 \end{bmatrix}_2.$$

The best one-gauge coherence fraction tends to $21/64$, while a random global potential explains on average $1/8$ of local potentials. The best one-gauge residual frustration tends to $43/64$.

8.3 Gluing kernel

For two three-planes P, Q with $r = \dim(P \cap Q)$, the compatibility probability of local potentials is

$$K(0) = 1, \quad K(1) = \frac{101}{155}, \quad K(2) = \frac{3}{5}, \quad K(3) = 1.$$

The corresponding joint matching probabilities for a random global gauge are

$$J_0 = \frac{1}{64}, \quad J_1 = \frac{101}{4960}, \quad J_2 = \frac{3}{80}, \quad J_3 = \frac{1}{8}.$$

8.4 First-Zero Coherence Law

Write a global gauge as bits

$$\lambda = (\lambda_1, \dots, \lambda_d),$$

scanning from the newest top coordinate downward. The explained-face count $N_d(\lambda)$ is determined by the first zero among $\lambda_d, \dots, \lambda_4$ and whether the lower tail $\lambda_1, \dots, \lambda_{k-1}$ is zero. If the first zero is at k , then

$$N_d(\lambda) = \begin{cases} 8^{d-k} A_k, & \text{if the lower tail is zero,} \\ 8^{d-k} B_k, & \text{if the lower tail is nonzero.} \end{cases}$$

If no zero appears above coordinate 3, then

$$N_d(\lambda) = \begin{cases} 8^{d-3}, & \text{if the lower tail is zero,} \\ 0, & \text{if the lower tail is nonzero.} \end{cases}$$

This yields exactly two coherence-scar gauges of maximal coverage and exactly seven dead gauges in every dimension.

8.5 Exact atlas theorem

Define an atlas as a set of global gauges covering every local three-face potential. The atlas number is the minimum number of gauges needed:

$$\tau_d = \min\{|\mathcal{A}| : \text{every three-face } P \text{ is covered by some } \lambda \in \mathcal{A}\}.$$

The exact atlas theorem is

$$\tau_d = d - 2.$$

A canonical minimal atlas is

$$\mathcal{A}_d = \{0, e_d, e_d + e_{d-1}, \dots, e_d + e_{d-1} + \dots + e_4\}.$$

It satisfies the recursion

$$\mathcal{A}_d = \{0\} \cup (e_d + \mathcal{A}_{d-1}).$$

The proof uses vertical faces, graph faces, and the gauge law $\theta_{\text{graph}(\ell)} = \theta_P + \ell$. A lower-bound argument shows fewer than $d - 2$ gauges must leave a graph lift uncovered.

Status. The upper bound is proved by recursion. A lower-bound proof sketch was developed and matches exact searches through $d = 7$. It should be formalized carefully before publication.

9 Associator Compression Theory and Marginal Chart Laws

9.1 Atlas intersection law

For the canonical atlas $\mathcal{A}_d = \{g_0, \dots, g_{d-3}\}$, let $C_d(S)$ be the number of three-faces covered by exactly the chart subset S and by no other chart. For every nonempty S ,

$$C_d(S) = 7^{\max(S) - |S| + 1} \begin{bmatrix} d - 1 - \max(S) \\ 2 \end{bmatrix}_2.$$

Every nonempty chart subset occurs, so the atlas nerve is the full simplex. This gives an exact chart-intersection geometry.

9.2 Exact and approximate atlas dichotomy

Exact coherence needs $d - 2$ charts. But if r structured suffix charts are used, the exact coverage is

$$\text{Cov}_{d,r} = \sum_{j=2}^{r+1} 8^{d-1-j} \begin{bmatrix} j \\ 2 \end{bmatrix}_2.$$

The asymptotic coverage fraction is

$$F_r = 1 - \left(\frac{7}{2^{r+2}} - \frac{14}{4^{r+2}} + \frac{1}{8^{r+1}} \right).$$

Charts r	Structured coverage F_r	Random coverage limit $1 - (7/8)^r$
1	0.328125	0.125000
2	0.615234	0.234375

3	0.794678	0.330078
4	0.894012	0.413818
5	0.946163	0.487091
6	0.972869	0.551205
8	0.993177	0.656391

Thus exact associator coherence is linearly complex, but approximate coherence is logarithmically cheap when charts are chosen structurally:

$$\tau_{\text{exact}}(d) = d - 2,$$

$$\tau_{\text{approx}}(\varepsilon) \approx \left\lceil \log_2 \left(\frac{7}{4\varepsilon} \right) \right\rceil.$$

9.3 Marginal Chart Law

The newest calculation refined the compression curve by identifying the marginal contribution of the r -th suffix chart. Let Δ_r be the asymptotic fraction of all local associator potentials newly covered by adding the r -th structured chart. Then

$$\Delta_r = F_r - F_{r-1} = \frac{21 \begin{bmatrix} r+1 \\ 2 \end{bmatrix}_2}{8^{r+1}}, \quad r \geq 1,$$

with $F_0 = 0$. The residual after r suffix charts is

$$E_r = 1 - F_r = \frac{7}{2^{r+2}} - \frac{14}{4^{r+2}} + \frac{1}{8^{r+1}}.$$

Therefore each additional structured chart approximately halves the remaining residual, with precise q -binomial corrections. This gives a repair half-life law for associator coherence.

r	Δ_r	Residual E_r
1	$21/64 \approx 0.328125$	$43/64 \approx 0.671875$
2	$147/512 \approx 0.287109$	$197/512 \approx 0.384766$
3	$735/4096 \approx 0.179443$	$841/4096 \approx 0.205322$
4	$3255/32768 \approx 0.099335$	$3473/32768 \approx 0.105988$
5	$13671/262144 \approx 0.052151$	$14113/262144 \approx 0.053837$
6	$56007/2097152 \approx 0.026706$	$56897/2097152 \approx 0.027131$

New useful law. Structured associator charts have a dyadic residual half-life: after the first few charts, each additional chart cuts the remaining local incoherence by about one half. Random charts do not have this behavior.

10 Computational Methods and Verification Status

The study used Python for finite enumeration, exact table checking, branch-and-bound semigroup repair search, Boolean polynomial analysis, subspace enumeration over $\mathbb{F}_2$, and formula verification. The major computational patterns were cross-checked in multiple ways where possible.

Task	Method	Verification status
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Count $\text{ns}(A)$	Enumerate all triples and compare $(xy)z$ with $x(yz)$	Exact for finite tables tested
Search $\text{sdist}(A)$	Branch on failed triples; final repair must change one of uv , $(uv)w$, vw , $u(vw)$	Exact through stated radii for G and A
Cayley–Dickson Φ_n	Recursive signed-basis multiplication over $\mathbb{F}_2^n$	Exact enumeration for low n ; recurrence checks higher n
Three-plane potentials	Enumerate ordered bases of $\mathbb{F}_2$ three-planes; solve for linear θ_P	Exhaustive for $n \leq 6$, random larger checks
Holonomy H_d	Enumerate global gauges and count matched three-faces; verify recurrence	Exact for small d ; recurrence checked through $d = 60$
Atlas theorem	Canonical cover recursion and exact search for lower bounds through $d = 7$	Proof sketch plus exact low-dimensional search
Compression laws	Use atlas-intersection formula and compare to random baseline	Formula verified computationally through tested ranges

Representative pseudocode.

```

for x in A:
    for y in A:
        for z in A:
            if op[op[x,y], z] != op[x, op[y,z]]:
                ns += 1

# Repair branch rule:
# if triple (x,y,z) fails, every final associative repair must change
# at least one of (x,y), (op[x,y],z), (y,z), (x,op[y,z]).
    
```

Caution. Many results here are preliminary findings, not peer-reviewed results. This document distinguishes proof-level statements, computational certifications, and conjectural laws.

11 Applications and Useful Interpretations

Area	Use of the theory
Software repair	A single failing law test may require multiple internal edits, or may indicate a missing state, class, or exception case.
Distributed systems	Associative merge laws matter for replicated data. Associator diagnostics can distinguish local merge bugs from missing metadata or coherence regimes.
AI and learned composition	Learned memory, plan, tool-output, or embedding composition may fail associativity because of bad rules, insufficient latent state, or missing coherence/provenance data.
Tensor networks and scientific computing	Order-dependence of approximate contractions may be repaired by increasing hidden dimension, tracking coherence corrections, or using a small atlas of correction charts.
Physics	Nonassociativity may signal missing degrees of freedom or genuine flux/cocycle curvature; finite associator sheaves provide toy models of higher-gauge obstruction.

Error-correcting algebra	Associativity constraints can be treated as syndrome checks;
	cohomological repair distance becomes a code-distance-like invariant.
Automata/state abstraction	A one-bug associator may indicate two states were wrongly merged;
	hidden-state repair is a state-splitting operation.

The practical diagnostic pipeline proposed by the theory is

detect associator field $\longrightarrow$ diagnose source $\longrightarrow$ fit local potentials $\longrightarrow$ test overlap compatibility
 $\longrightarrow$ search coherence scars $\longrightarrow$ build structured atlas $\longrightarrow$ patch only the residual.

12 Open Problems and Future Research Targets

1. Finite density problem: can one-bug n -element magmas have semigroup distance $\Omega(n^2)$, or is $O(n)$ optimal?
2. Hidden-state resistance: find one-bug magmas with large internal semigroup distance and large completion number, not collapsible by one hidden state.
3. Formalize the eight-element distance-5 table as a theorem, preferably with a compact certificate rather than exhaustive search.
4. Prove the Cayley–Dickson curvature spectrum law for all n .
5. Prove the three-plane potential law and four-space/global holonomy laws formally from the Cayley–Dickson recurrence.
6. Complete the all-gauge optimality proof for suffix atlases in the budgeted coverage problem.
7. Develop algorithms for associator source separation in real finite systems, especially distinguishing projection curvature from cocycle curvature.
8. Explore approximate atlas construction in learned systems and distributed merge protocols.
9. Compute exact cohomological repair distance $d_{\text{cocyc}}(n)$ for $n \geq 5$.
10. Study whether associator atlas geometry has a cohomological interpretation as a finite sheaf/gerbe invariant.

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